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The Editors

Dear Editors,

I am pleased to submit for consideration the manuscript "**Correlation Rigidity and Contagion in Stablecoin Baskets: A Structural Diagnostic for Supervisors**."

US-dollar stablecoins now settle a systemic share of crypto transactions, and three institutional arrangements price the same object: basket margin at centralized venues, index perpetuals on decentralized venues, and proof-of-reserves at regulated issuers. All three need the joint peg-deviation correlation surface of a basket. Nobody observes that surface. On a representative day only three or four coins deviate at all, and the correlation matrices that clearinghouses, index providers and regulators publish are smoothed across the gaps. The paper's starting observation is that the gaps are not statistical noise. They are structural discontinuities, and smoothing across them systematically understates the joint tail loss the exercise is meant to price.

The paper turns that observation into a diagnostic a supervisor can actually run. It represents the joint peg-preserving equilibrium as a parameter space stratified by the boundaries separating distinct failure channels—backing ratio, oracle, redemption, custody, bridge, regulatory, funding rate and interest rate—so that the stratification is inherited from the peg constraint rather than estimated from data. Within a stratum correlations are locally constant. Across a boundary they jump, and the jump is a rank-one object supported on a small block of coins, with a sign that the taxonomy fixes before any event window is observed.

Three things follow, and they are what I would ask a referee to judge.

1. **An obstruction, not a fitting exercise.** The paper proves a matching pair of bounds on correlation recovery: a local-constancy result from below, and a closed-form rank-one residue from above. They meet at an intrinsic loss, which says that every locally continuous estimator—Gaussian copula, single-factor portfolio invariant, low-rank shrinkage, dynamic conditional correlation—must forgo the jump. That is a statement about the class of admissible estimators, not about one sample.
2. **An out-of-sample test with the sign fixed in advance.** The amplitude is calibrated once, on the USDC-SVB custody event of March 2023, and then held fixed. The block prediction beats the Gaussian copula across three metric families under a stationary block bootstrap, and the head-to-head sign transports to a structurally distinct second event—FTX contagion in November 2022—without refitting. The two boundary cases are disclosed rather than absorbed: FDUSD-2024 falls outside the calibrated failure family, and the UST/LUNA collapse exits the regime in which the first-order approximation is valid. Two conditional placebo tests do not reject the null and are reported as boundaries of the identifying claim. No aggregate distributional-inference power is asserted.
3. **A diagnostic computable from public data.** Every input is a Level-1 public source: the

DeFiLlama peg-price archive, FRED series, Yahoo Finance and Etherscan records. Nothing proprietary, paywalled or issuer-internal. This matters operationally: the institutions that would use the diagnostic do not publish liquid equity prices and do not disclose bilateral exposures, which is precisely why equity-based systemic-risk measures have no traction here. A decomposition of basket tail loss then separates a structural correlation channel from a mechanical volatility channel that any acute-window estimator would pick up. Three policy directions follow, all falsifiable on the public panel shipped with the paper: a block-alignment audit statistic, a boundary-type-conditioned margin multiplier, and a public-book pooling protocol.

The reproducibility bundle contains all source, figure-generation and audit scripts, and raw extracts with SHA-256 checksums; the pipeline completes in under three minutes.

The manuscript is the author's original work, has not been published previously, and is not under consideration elsewhere. It contains no material requiring copyright permission. The author has no conflicts of interest to disclose.

Thank you for considering the manuscript.

Sincerely,

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